from waves of migration, from Europe, but also material economic and military resources from France and other European powers. Before and after World War II, an influx of Jewish and German scientists contributed decisively to an American technological edge in its global competition with the USSR. [19] Meanwhile, Russia advanced tremendously thanks to external assistance in its rapid domestic transformation into a European and global power after the seventeenth century. This included Peter the Great's import of shipbuilding and other technologies from western Europe and extended to the Soviet Union's use of German scientific capabilities after World War II. [20]

But for India, more illustrative examples of domestic transformation come from Asia. Japan, for example, had remained an isolated, feudal society until its leaders suddenly discovered the modern advancements made by the United States and Europe in the 1850s and 1860s. Soon after the Meiji Restoration, between 1871 and 1873, a group of almost 100 Japanese led by the nobleman Iwakura Tomomi went on a round-the-world mission. The Iwakura Mission's primary purpose was diplomatic, to revise what Japan saw as unequal treaties with other countries. But it was also a tour of study: it was meant to discover what had made the West so successful. It resulted in a discovery of education systems, industrial centres, and dockyards in the United States, Europe and elsewhere. Students accompanying the mission were left to study overseas, with the objective of returning to accelerate the process of Japan's modernization. [21]

The modernization effort that followed in Japan was swift, but more difficult and painful than many often appreciate. It was financed initially by silk and tea exports overseas, particularly to a rising United States. The government faced fiscal constraints, having had to compensate feudal landowners. Efforts to modernize the Japanese military, using conscription, led to riots in what had to date been a hierarchical, casteist, warrior society. Private universities arose to rival Japan's state-supported institutions but were often criticized as hotbeds of radicalism. Industrialization did come about eventually, but only in the late 1880s and early 1890s. It was a full three decades after the Iwakura Mission that Japan announced itself on the